

Lab 4

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Questions

1) Increasing “i_focal_dist” makes the image appear zoomed in, so the objects look larger and less of the scene is visible. Decreasing it makes the image appear zoomed out, so the objects look smaller and more of the scene becomes visible. This happens because a larger focal distance makes the rays more parallel to the camera’s viewing direction, giving a narrower field of view. A smaller focal distance spreads the rays farther apart and gives a wider field of view.

2) Super-sampling could be implemented by shooting several rays through slightly different positions inside each pixel instead of shooting only one ray through its centre. The positions could follow a small regular grid or use random offsets. The colours returned by all the rays would then be added together and divided by the number of samples. Pixels close to sphere edges or checkerboard boundaries would contain an average of both sides, producing smoother edges and reducing aliasing. More samples improve the result but also increase the rendering time.

3) “MAX_DEPTH” controls how many primary and secondary rays can be processed for each pixel. At depth 1, a refractive object mostly appears dark or opaque because the ray stops at its front surface. At depth 2, the ray can enter the object but cannot fully continue through the back surface, so the result is still incomplete. At depth 3, the ray can enter, leave and reach the scene behind the object, making refraction visible.

Reflective objects need fewer steps. The first depth shows the surface itself, depth 2 shows the first reflection, and depth 3 allows reflections inside other reflections.

In the lab scene, changes become very small after around depth 4 or 5 because most rays have reached the sky or a non-reflective surface, while the remaining ray weights are too small to noticeably affect the image.